📊 CUSTOMER CHURN ANALYSIS REPORT

🔹 1. Project Overview

This project focuses on analyzing customer churn data to understand customer behavior, identify patterns in churn, and provide actionable business insights. The analysis is performed using Python and Pandas for data manipulation and visualization.

🔹 2. Objectives

- Identify customers who are likely to churn
- Analyze factors affecting churn
- Understand revenue contribution
- Provide recommendations to improve customer retention

🔹 3. Dataset Description

The dataset contains customer-related information including:

- Customer ID
- Tenure (number of months the customer stayed)
- Monthly Charges
- Total Charges
- Contract Type
- Payment Method
- Paperless Billing
- Senior Citizen
- Churn Status

🔹 4. Data Cleaning & Preparation

- Converted **TotalCharges** column to numeric format
- Handled missing values using appropriate techniques
- Checked for duplicate records
- Ensured correct data types for analysis

🔹 5. Data Analysis & Techniques Used

✓ Aggregation

- Total Revenue calculated using sum of TotalCharges
- Average Monthly Charges calculated using mean
- Total Customers counted using unique values

✓ Grouping

- Churn rate analyzed by Contract type
- Revenue analyzed by Payment Method

✓ Filtering

- Identified high-risk customers using multiple conditions

✓ Pivot Tables

- Created pivot tables to summarize customer behavior and charges

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🔹 6. Key Metrics

- **Total Customers:** 500
- **Total Revenue:** 2118941
- **Average Monthly Charges:** 113.636

🔹 7. Key Insights

- Customers with **higher monthly charges** tend to churn more frequently
- Customers with **month-to-month contracts** have the highest churn rate
- Long-term contract customers show **better retention**
- Certain payment methods contribute more to total revenue
- Customers with low tenure are more likely to churn

🔹 8. Visualizations

The following charts were created to support analysis:

- Churn Distribution (Bar Chart)
- Monthly Charges vs Churn (Bar Chart)
- Contract Type vs Churn Rate (Bar Chart)

- Revenue by Payment Method (Bar Chart)

🔹 9. Business Recommendations

- Offer incentives for customers to choose long-term contracts
- Provide special offers to high-paying customers
- Improve customer engagement strategies
- Focus on retaining new customers (low tenure group)
- Enhance service quality to reduce churn

🔹 10. Conclusion

This analysis provides valuable insights into customer churn behavior. By focusing on high-risk customers and improving retention strategies, businesses can reduce churn and increase overall profitability.

🔹 11. Tools & Technologies Used

- Python
- Pandas
- Matplotlib
- NumPy

🔹 12. Future Improvements

- Use machine learning models to predict churn
- Build an interactive dashboard using Tableau or Power BI
- Perform deeper segmentation analysis